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# a) Stimulating Static Routing Configuration

## Aim:

To simulate and configure static routing using Cisco Packet Tracer.

## Static Routing:

Static Routing is a method of routing in which the routes are manually configured by the network administrator. The administrator specifies the destination network, subnet mask and ~~network~~ next-hop address. In static routing, the router does not automatically learn routes from the routers. Therefore, the administrator must configure the required routes manually.

Static routers can also be configured with different Administrative Distance. A route with lower administrative distance is preferred over a route with a higher administrative distance.

Procedure:

i) Create the required network topology in Cisco Packet Tracer.

ii) Connect the ~~required~~ ~~nets~~ devices using appropriate cables.

iii) Configure the IP Addresses and subnet masks for all PCs and routers.

iv) Enable router interfaces and configure static routes in each ~~root~~ router.

v) Configure a primary route with the Lower Administrative Distance.

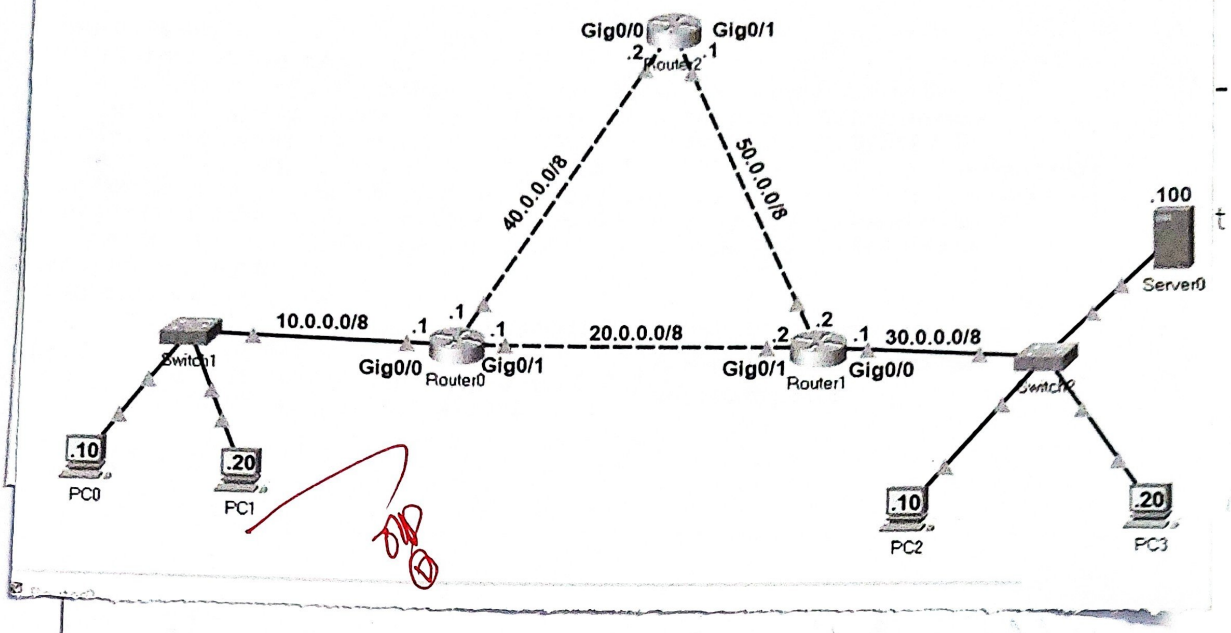
vi) Configure a backup route with a higher Administrative Distance.

vii) Display the static routing cable using,

# show ip route static

viii) Test the connectivity between PCs ~~and~~ using ping. Then check path followed by packets using tracert.

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| Device |
|---|
| PC0 |
| Router0 |
| Router0 |
| Router0 |
| Router1 |
| Router1 |
| Router2 |
| Router2 |
| Router2 |
| PC1 |

Result:

The simulation of Static Routing has been implemented successfully.

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# b) Simulation of RIP

## Aim:

To simulate RIP using Cisco Packet Tracer.

## Initial IP Configuration

| Device | Interface | IP Configuration | Connected with |
|---|---|---|---|
| PC0 | Fast Ethernet | 10.0.0.2/8 | Router0's Fa0/1 |
| Router0 | Fa0/1 | 10.0.0.1/8 | PC0's Fast Ethernet |
| Router0 | S0/0/1 | 192.168.1.254/30 | Router2's S0/0/1 |
| Router0 | S0/0/0 | 192.168.1.249/30 | Router1's S0/0/0 |
| Router1 | S0/0/0 | 192.168.1.250/30 | Router0's S0/0/0 |
| Router1 | S0/0/1 | 192.168.1.246/30 | Router2's S0/0/0 |
| Router2 | S0/0/0 | 192.168.1.245/30 | Router1's S0/0/1 |
| Router2 | S0/0/1 | 192.168.1.253/30 | Router0's S0/0/1 |
| Router2 | Fa0/1 | 20.0.0.1/30 | PC1's Fast Ethernet |
| PC1 | Fast Ethernet | 20.0.0.2/30 | Router2's Fa0/1 |

## Routing Information Protocol (RIP)

Routing Information Protocol is a dynamic routing protocol used by routers to exchange information about the networks connected to them. Unlike static routing, routes do not have to be manually entered to every destination network. The routers exchange routing information and

learn the available paths automatically. RIP uses hop count as its routing metric.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
```

```
Router(config-if)#exit
Router(config)#interface serial 0/0/1
Router(config-if)#ip address 192.168.1.254 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#bandwidth 64
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
```

**Command Prompt**

```
Packet Tracer PC Command Line 1.0
PC>ipconfig

FastEthernet0 Connection:(default port)
Link-local IPv6 Address.....: FE80::260:70FE
IP Address.....: 20.0.0.2
Subnet Mask.....: 255.0.0.0
Default Gateway.....: 20.0.0.1

PC>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Request timed out.
Reply from 10.0.0.2: bytes=32 time=3ms TTL=126
Reply from 10.0.0.2: bytes=32 time=3ms TTL=126
Reply from 10.0.0.2: bytes=32 time=3ms TTL=126

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 3ms, Average = 3ms

PC>
```

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Procedure :

i) Create the topology using three routers and two PCs in the Cisco Packet Tracer.

ii) Connect the routers and PCs. Then connect routers using serial links.

iii) Configure the IP Addresses and subnets masks for all PCs and router interface.

iv) Enable all router interfaces using #no shutdown.

v) Configure the clock rate on the DCE serial interface.

vi) Configure RIP on each router and allow the routers to exchange routing information.

vii) Check connectivity and use tracert to observe the route followed by the packets.

Result :

The Simulation of RIP using Cisco Packet Tracer has been completed.